LECTURE 2

2. Splitting fields and normal extensions

Definition 2.1. An extension $L \supseteq K$ is a *splitting field* for a polynomial $f \in K[t]$ if

(a) f equals a product of linear factors in $L[t]$ (*splits* over L): $f = k(t-a_1)\cdots(t-a_n)$ for $a_i \in L$ (with $k \in K$, or $k = 1$ if f is monic). In other words L contains all roots a_i of f; and

(b) L is minimal possible: if $K \subseteq L_1 \subseteq L$ is such that f also splits over L_1, then $L_1 = L$.

Obviously, these two conditions are equivalent to $L = K(a_1, \ldots, a_n)$, where the a_i are all roots of f.

Remark 2.2. 1. Certainly, the roots of f do not change if we multiply f by a non-zero element of K. Thus for splitting fields we may always regard f as being monic.

2. Suppose L is a splitting field for $f \in K[t]$ and V is an intermediate field: $K \subseteq V \subseteq L$. Then $f \in V[t]$ too, and L is also a splitting field for f over V. Indeed, the polynomial f still splits over L and $L = V(a_1, \ldots, a_n)$, where the a_i are the roots of f.

Example 2.3. A splitting field for $t^3 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2})$, where $\omega = e^{2\pi i/3}$ is a root of $t^2 + t + 1$ (being a complex root of $t^3 - 1 = (t-1)(t^2+t+1)$). One can show that $\mathbb{Q}(\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}) = \mathbb{Q}(\sqrt[3]{2}, \omega)$ (see Exercise Sheet 1). By the Tower Law applied to $\mathbb{Q}(\sqrt[3]{2}, \omega) \supseteq \mathbb{Q}(\sqrt[3]{2}) \supseteq \mathbb{Q}$ we see that the resulting extension is of degree 6: the degree of ω over $\mathbb{Q}(\sqrt[3]{2})$ is 2 since ω is a root of a quadratic, and it is not of degree 1 since ω is not real, and the degree of $\sqrt[3]{2}$ over $\mathbb{Q}$ is 3 since $t^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's.

Theorem 2.4. (Existence of splitting fields) *For every $f \in K[t]$ over any field K there is a splitting field $L \supseteq K$ for f.*

Proof. The proof is by induction on $\deg f$.

If $\deg f = 1$, the (only) root of f is in K since f is linear. So put $L = K$.

If $\deg f > 1$, decompose f into a product of irreducible polynomials. Choosing one of the irreducible factors, we may write $f = mh$, where m is a (monic) irreducible polynomial (possibly $m = f$). By the theorem on the existence of simple extensions, there is a field $K(a)$ with m being the minimal polynomial of a. Then $f(a) = m(a)h(a) = 0 \cdot h(a) = 0$ so a is a root of f. Then $f = (t-a)g$ for some $g \in K(a)[t]$. We have $\deg g = \deg f - 1$. By the induction hypothesis applied to $K(a)$ and g, there is a splitting field L for g over $K(a)$.

Claim: L is the required splitting field for f over K. Indeed, the polynomial g splits over L, so $g = (t-a_2)\cdots(t-a_n)$ for $a_i \in L$. Then $f = (t-a)(t-a_2)\cdots(t-a_n)$ so f splits over L. We have $L = K(a)(a_2, \ldots, a_n) = K(a, a_2, \ldots, a_n)$, so L is generated by K and the roots of f. Thus L is a splitting field for f over K. □

Theorem 2.5. (Isomorphisms of splitting fields) *Suppose*

- *$i : K \to K'$ is a field isomorphism;*
- *$L \supseteq K$ is a splitting field for (monic) $f \in K[t]$; and*
- *$L' \supseteq K'$ is a splitting field for $i(f) \in K'[t]$.*

Then there exists an isomorphism $\mu : L \to L'$ such that $\mu|_K = i$.

Proof. We proceed by induction on $\deg f$. The idea is to use the theorem on isomorphisms of simple extensions, constructing the required isomorphism μ step-by-step, adjoining one root at a time.

If $\deg f = 1$ then all roots of f (of $i(f)$) are in K (in K') so $L = K$ (and $L' = K'$) and we take $\mu = i$.

Now let $\deg f > 1$. Using the decomposition of f into a product of irreducibles, we may write $f = mh$ for some irreducible $m \in K[t]$ (possibly $m = f$). Since L is a splitting field for f, we have $mh = f = (t - a_1) \cdots (t - a_n)$ for the roots $a_i \in L$. By the uniqueness of the decomposition, one of the factors on the right divides m. Without loss of generality, we may assume that $(t - a_1) \mid m$ and hence $m(a_1) = 0$. Since m is irreducible, we have m is the minimal polynomial of a_1 over K.

Since i is an isomorphism, we also have $i(f) = i(m)i(h)$ with $i(m)$ irreducible over K'. Since L' is a splitting field for $i(f)$, we have $i(m)i(h) = i(f) = (t-b_1) \cdots (t-b_n)$ for the roots $b_i \in L'$. By the uniqueness of the decomposition, one of the factors on the right divides $i(m)$. Without loss of generality, we may assume that $(t-b_1) \mid i(m)$ and hence $i(m)(b_1) = 0$. Since $i(m)$ is irreducible, we have $i(m)$ is the minimal polynomial of b_1 over K'.

Now, by the theorem on isomorphisms of simple extensions, there is an isomorphism $\lambda : K(a_1) \to K'(b_1)$ such that $\lambda(a_1) = b_1$ and $\lambda|_K = i$.

Since $f(a_1) = 0$, the polynomial $t - a_1$ divides f over $K(a_1)$: i.e. $f = (t - a_1)g$ for some $g \in K(a_1)[t]$. Of course, we have $g = (t - a_2) \cdots (t - a_n)$ over L and $L = K(a_1, a_2, \ldots, a_n) = K(a_1)(a_2, \ldots, a_n)$. By definition, the field L is a splitting field for g over $K(a_1)$.

Applying λ we obtain $\lambda(f) = (t - b_1)\lambda(g)$. Since $\lambda|_K = i$ and $f \in K[t]$, the LHS is $i(f)$. Then $\lambda(g) = (t - b_2) \cdots (t - b_n)$ over L' and $L' = K'(b_1, b_2, \ldots, b_n) = K'(b_1)(b_2, \ldots, b_n)$. Similarly, the field L' is a splitting field for $\lambda(g)$ over $K'(b_1)$.

We have $\deg g = \deg f - 1$. Hence by the induction hypothesis applied to $\lambda : K(a_1) \to K'(b_1)$ instead of i and to g instead of f, we obtain an isomorphism $\mu : L \to L'$ such that $\mu|_{K(a_1)} = \lambda$. Then we have $\mu|_K = \lambda|_K = i$, so μ is the required isomorphism. □

We remark that even if $K = K'$, given a polynomial $f \in K[t]$, there can be two distinct (but of course isomorphic) splitting fields for f over K. For example, the polynomial $t^2 - 2 \in \mathbb{Q}[t]$ has $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}[x]/(x^2 - 2)$ as splitting fields. Similarly the polynomial $t^2 + 1 \in \mathbb{R}[t]$ has the following three fields as possible splitting fields: $\mathbb{C}$, $\mathbb{R}[x]/(x^2 + 1)$, $\mathbb{R}[y]/(y^2 + 2y + 2)$.

Definition 2.6. A *monomorphism* is a homomorphism which is injective (that is, it is an isomorphism onto the image). We say that φ is a *K-monomorphism* if furthermore $\varphi|_K = \varepsilon_K$, the identity mapping of K. Likewise φ is a *K-homomorphism* if φ is a homomorphism and $\varphi|_K = \varepsilon_K$.

Example 2.7. The $\mathbb{Q}$-homomorphisms $\alpha : \mathbb{Q}(t) \to \mathbb{Q}(t)$ and $\beta : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{2})$, respectively defined by $t \mapsto t^2$ and $\sqrt{2} \mapsto -\sqrt{2}$ (and all elements of $\mathbb{Q}$ are fixed), are $\mathbb{Q}$-monomorphisms. On the other hand, the map $\gamma : \mathbb{F}_p[t] \to \mathbb{F}_p[t]$ defined by $f(t) \mapsto f(0)$ is an $\mathbb{F}_p$-homomorphism but not an $\mathbb{F}_p$-monomorphism.

Corollary 2.8. (Extension of monomorphisms to splitting fields) *Suppose that W is a splitting field for $f \in U[t]$ and suppose that $\lambda : V \to W$ is a U-monomorphism of V into W for some intermediate field $U \subseteq V \subseteq W$. Then λ can be extended to an isomorphism of W. That is, there exists an isomorphism $\mu : W \to W$ such that $\mu|_V = \lambda$.*

Proof. From Remark 2.2(2), we have W is a splitting field for f over V, and also W is a splitting field for f over $\lambda(V)$. Note that $\lambda(f) = f$ since $\lambda|_U = \varepsilon_U$ and $f \in U[t]$. So we are in

a special case of the theorem on isomorphisms of splitting fields, where $L = L' = W$, $K = V$, $K' = \lambda(V)$, $i = \lambda$. By that theorem, there exists an isomorphism $\mu : W \to W$ such that $\mu|_V = \lambda$, as required. □

Definition 2.9. An extension $L : K$ is *normal* if every irreducible polynomial $g \in K[t]$ that has a root in L splits over L.

Example 2.10. 1. $\mathbb{C} : \mathbb{R}$ and $\mathbb{C} : \mathbb{Q}$ are normal, since all polynomials split over $\mathbb{C}$.

2. The extension $\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}$ is not normal: the irreducible polynomial $t^3 - 2$ has the root $\sqrt[3]{2}$ in $\mathbb{Q}(\sqrt[3]{2})$, but it does not split over $\mathbb{Q}(\sqrt[3]{2})$ since the other two of its roots are not real.

3. We shall see later that $\mathbb{Q}(\sqrt{2}, i)$ is a normal extension. At first glance, it seems rather difficult to check this: we need apparently to check that the minimal polynomial of every element of $\mathbb{Q}(\sqrt{2}, i)$ splits over this field. However we shall now prove a very useful criterion.

Theorem 2.11. (Normal = splitting) *An extension $L : K$ is finite and normal if and only if L is a splitting field for some polynomial $f \in K[t]$.*

Proof. Suppose that $L : K$ is finite and normal. Then, by the theorem on finite extensions, $L = K(a_1, \ldots, a_s)$ for some a_i algebraic over K. Let $m_i \in K[t]$ be the minimal polynomial of a_i and put $f = m_1 \cdots m_s$. Now L is a splitting field for f; see Exercise Sheet 3.

Now suppose that L is a splitting field for $f \in K[t]$. First we show that $L : K$ is finite. We have $L = K(a_1, \ldots, a_n)$, where $\{a_1, \ldots, a_n\}$ is the set of all roots of f. Clearly, each a_i is algebraic over K (being a root of f). Then $L : K$ is finite by the theorem on finite extensions (part (b)).

To show that $L : K$ is normal, we need to show that if an irreducible polynomial $g \in K[t]$ has a root $\beta \in L$ (so $g(\beta) = 0$), then g splits over L. Let M be a splitting field for g over L, so that $K \subseteq L \subseteq M$. Then in fact M is a splitting field for fg over K: indeed, clearly fg splits over M and M is generated by L and the roots of g, and hence by K and the roots of f and the roots of g (since L is generated by K and the roots of f). We need to show that every root $\gamma \in M$ of g belongs to L. (This will, of course, imply that $M = L$.)

By the theorem on isomorphisms of simple extensions, there is an isomorphism $\lambda : K(\beta) \to K(\gamma)$ such that $\lambda(\beta) = \gamma$ and $\lambda|_K = \varepsilon_K$. So λ is a K-monomorphism of $K(\beta)$ into M (which, recall, is a splitting field for fg over K). By the corollary on extending monomorphisms to splitting fields, there exists an isomorphism $\mu : M \to M$ such that $\mu|_{K(\beta)} = \lambda$.

We claim that $\mu(L) = L$. Indeed, for every root a_i of f we have

$$f(a_i) = 0 \qquad \Longrightarrow \qquad \mu(f(a_i)) = \mu(0) = 0.$$

However $\mu|_K = \lambda|_K = \varepsilon_K$, hence $\mu(f(a_i)) = f(\mu(a_i))$ since μ is an isomorphism. So μ maps any root of f to a root of f. Since μ is injective, we have $\{\mu(a_1), \ldots, \mu(a_n)\} = \{a_1, \ldots, a_n\}$ (irrespective of possible multiplicities). Now

$$\mu(L) = \mu(K(a_1, \ldots, a_n)) = \mu(K)(\mu(a_1), \ldots, \mu(a_n)) = K(a_1, \ldots, a_n) = L$$

since μ is an isomorphism.

Hence $\gamma = \lambda(\beta) = \mu(\beta) \in \mu(L) = L$, as required. □

Example 2.12. We can now prove that $\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}$ is a normal extension: simply because this is a splitting field for $(t^2 + 1)(t^2 - 2)$.